

Infosys Springboard

Virtual Internship 6.0

Milestone 1– Requirement Gathering & Planning

Infosys ElectViz Election Data Visualization on for Media

Project Title: 2024 Indian General Election

Platform: Power BI

Dataset Source: Kaggle – Lok Sabha Election 2024 Dataset
(<https://www.kaggle.com/datasets/rubenmukherjee/india-2024-election-dataset-all-candidates>)

Presented By: Shruti Bhore

1.Requirement Analysis

- Understood the client requirement to build a Power BI Election Analytics Dashboard for Lok Sabha Elections 2024.
- **Identified the key objectives:**
 - Display national-level election summary
 - Show alliance-wise & party-wise comparisons
 - Provide state-level and constituency-level drill-down
 - Enable map-based geographic insights
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- **Analyzed what metrics are required:**
 - Seats Won, Vote Share, Alliances, Margin, Total Votes, State-wise breakup
- Confirmed requirement of at least 3–4 interactive dashboard pages.

2.Abstract

This project focuses on analyzing and visualizing the 2024 Indian General Election using Microsoft Power BI. Raw election data was cleaned, transformed, and modelled to generate meaningful and interactive dashboards. The dashboards clearly highlight alliance performance, party-wise seat distribution, and state-level voting patterns. The objective is to convert complex election data into an easy-to-understand visual format that supports both academic study and public understanding.

3.Backlog Creation

Created an initial product backlog for tasks needed to complete the project:

Product Backlog Items

1. Dataset identification from Kaggle
2. Understanding dataset columns and importance
3. Data cleaning requirements list
4. Data model planning
5. Dashboard design planning
6. Creation of visuals required (bar, pie, maps, KPI cards)
7. Requirement documentation
8. Stakeholder review & changes tracking
9. Power BI themes, color selection, UI layout
10. Defining KPIs (Seats Won, Majority, Party Share)

4.Sprint Planning

Planned sprint structure for initial development:

- **Sprint 1 Focus**
 - Dataset collection
 - Understanding data & column mapping
 - Drafting KPIs
 - Creating product backlog
- **Sprint Goal**
 - To prepare the dataset and finalize dashboard requirements before starting design.

5.Dataset Identification

Dataset selected from **Kaggle – Lok Sabha Election 2024 Dataset**

Key dataset features:

- Contains national, state and constituency-level results
- Columns include:
 - State
 - Constituency
 - Winning Party
 - Winning Candidate
 - Total Votes
 - Alliance
 - Margin
 - Turnout %
- Dataset suitable for Power BI because of:
 - Clean structured format
 - Clear hierarchy (India → State → Constituency)
 - Contains metrics needed for visualization
 - Easily supports map visuals

6.Grooming

- Reviewed each backlog item with the team
- Prioritized tasks required for Milestone 2
- Identified which columns are essential and which are less important
- Finalized visualization needs:
 - Alliance cards
 - Party-wise seats
 - Map visuals
 - Drill-down charts

7.Sprint Backlog

Sprint backlog items confirmed for Milestone 1:

1. Analyze dataset
2. Verify missing/invalid values
3. Finalize key columns for visuals
4. Define required calculated fields (DAX)
5. Plan color theme and dashboard layout
6. Prepare requirement documentation
7. Conduct stand-up discussions (daily status updates)

8. Stand-up Summary

- Discussed daily progress
- Shared challenges: dataset cleaning, margin calculation
- Decided responsibilities for upcoming sprint
- Ensured smooth progress towards Milestone 2 deliverables

Conclusion

Milestone 1 successfully established a strong foundation for the Election Analytics Dashboard project. Through detailed requirement gathering, dataset identification, and structured planning, we clearly defined all functional needs, KPIs, and dashboard expectations. The team analyzed the Kaggle Lok Sabha Election 2024 dataset, evaluated its suitability for Power BI, and created a well-organized product backlog and sprint plan.

This milestone ensures that the project has a clear direction, a validated dataset, and a roadmap for development. With the planning phase completed, the team is now prepared to move confidently into Milestone 2, focusing on data cleaning, modeling, and dashboard development.

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Milestone 2 – Development Phase (Part 1)

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1. Overview of Milestone 2

Milestone 2 focuses on turning the election dataset into a structured model and building the first draft of the dashboard. This includes:

- ❑ **Data Modelling**
- ❑ **Visualization Design**
- ❑ **Page 1 & Page 2 Development**
- ❑ **Initial DAX Creation**

This milestone builds the foundation for a fully interactive and meaningful election analytics report.

2. Abstract

This project focuses on analyzing and visualizing the 2024 Indian General Election using Microsoft Power BI. Raw election data was cleaned, transformed, and modelled to generate meaningful and interactive dashboards. The dashboards clearly highlight alliance performance, party-wise seat distribution, and state-level voting patterns. The objective is to convert complex election data into an easy-to-understand visual format that supports both academic study and public understanding.

3. Objectives

The main objectives of the project are:

- To compare the overall performance of NDA, INDIA, and other alliances
- To analyze party-wise seat contribution at the national level
- To study state-wise and constituency-wise voting patterns
- To apply DAX measures for accurate calculation of seats, votes, and margins
- 2To present election results in a clear, interactive, and user-friendly format

4. Tools & Requirements Gathering

4.1 Tools Used

- Power BI Desktop – For data modelling, visual creation
- Power Query Editor – For data cleaning
- Kaggle Dataset – Election results data
- Excel – For reviewing and verifying dataset
- DAX – For calculated measures

4.2 Requirement Gathering

Based on the project scope and mentor instructions, the requirements for dashboard the were:

Business Requirements

1. Compare party performance across states.
2. Identify winning margins by party and candidate.
3. Show state-wise seat distribution using graphs & maps.
4. Provide visual-level filters (party, state, candidate).
5. Create clean, simple, professional visuals.

Technical Requirements

- Use at least 5+ visuals
- Use DAX measures
- Use data modelling (Star Schema)
- Provide at least two dashboard pages

5. Dataset Description

Dataset Name: India Lok Sabha Election 2024 – State-wise & Constituency-wise Results

Rows & Columns:

- **Rows:** ~540 (One per constituency)
- **Columns:**
 - *State*
 - *Constituency Name*
 - *Winning Candidate*
 - *Winning Party*
 - *Total Votes*
 - *Margin*
 - *Runner-Up Candidate*
 - *Runner-Up Party*
 - *Constituency Turnout %*
 - *Gender-wise Vote Split (if available)*

Column	Importance	Reason
State	High	For state-wise visualization & maps
Constituency	High	For drill-down analysis
Winning Party	High	To show party performance in India
Winning Candidate	Medium	Candidate level insights
Total Votes	High	Needed for charts & comparisons
Margin	Medium	Helps measure strong vs weak wins
Turnout %	Medium	Helps show voter participation rates

6.DAX Formulas Used

To convert raw election data into meaningful insights, several DAX measures and calculated fields were created:

- **Total Votes** – Calculates the total number of votes across all constituencies
- **Total Seats Won** – Counts the total number of constituencies won
- **Average Margin** – Measures the average victory gap to identify close and strong wins
- **Winning Candidate** – Identifies the candidate with the highest votes in each constituency
- **Winning Party** – Identifies the political party of the winning candidate
- **Winning Vote Share Percentage** – Displays the vote share of the winning candidate

These measures form the analytical backbone of the dashboard.

Total Votes $\text{Total Votes} = \text{SUM}(\text{Votes_Fact}[\text{TotalVotes}])$

Purpose: Calculates the total number of votes received across all constituencies for overall voting analysis.

Total Seats Won $\text{Total Seats Won} = \text{DISTINCTCOUNT}(\text{Votes_Fact}[\text{Constituency}])$

Purpose: Counts how many constituencies were won, which helps measure performance at party and state levels.

Avg Margin $\text{Avg Margin} = \text{AVERAGE}(\text{Votes_Fact}[\text{Margin}])$

Purpose: Computes the average winning margin to understand the strength of victories—whether the elections were closely contested or one-sided.

Winning Candidate

Winning Candidate = $\text{VAR MaxVotes} = \text{MAX}(\text{'ConstituencyWiseDetails'}[\text{Total Votes}])$

RETURN

$\text{CALCULATE}(\text{MAX}(\text{'ConstituencyWiseDetails'}[\text{Candidate}]), \text{'ConstituencyWiseDetails'}[\text{Total Votes}] = \text{MaxVotes})$

Purpose: Identifies the candidate who received the highest number of votes in the selected constituency.

Winning Party Name

Winning Party Name = $\text{VAR MaxVotes} = \text{MAX}(\text{'ConstituencyWiseDetails'}[\text{Total Votes}])$

RETURN

CALCULATE(MAX('ConstituencyWiseDetails'[Party]),
'ConstituencyWiseDetails'[Total Votes] = MaxVotes)

Purpose: Determines the political party of the winning candidate.

Winning Candidate Vote Share %

Winning Candidate Vote Share % =

"Vote Share: " & MAX('ConstituencyWiseDetails'[% of Votes]) & " %"

Purpose: Displays the vote share percentage of the winning candidate as a text label in visuals.

7. Data Modelling

Model Used: Star Schema

Fact Table:

✓ Votes_Fact (Total Votes, Margin, Turnout%)

Dimension Tables:

✓ Dim_State

✓ Dim_Constituency

✓ Dim_Party

✓ Dim_Candidate

Relationships Created:

- State → Constituency
- Constituency → Votes_Fact
- Party → Votes_Fact

Outcome:

A clean and optimized data model for faster calculations and accurate filtering.

8. Visualization Design

The visualization design for the India General Election Results – 2024 dashboard focuses on creating a clear, interactive, and user-friendly analytical experience. The design approach aligns with best practices of BI dashboards, ensuring accuracy, readability, and insight-driven storytelling. Each page was designed to answer specific business questions using relevant charts, maps, KPIs, and slicers.

8.1 Overall Design Principles Used

✓ Color Coding (Meaningful & Consistent)

- **Saffron/Orange** → NDA Alliance
- **Blue/Green** → INDIA Alliance
- **Grey/Blue** → Others Party
- **Map shades represent:**
 - **Dark color** = Higher seats
 - **Light color** = Lower seatsThis ensures instant recognition of alliances and performance across visuals.

✓ Layout Principles

- KPIs placed at the top for quick summary.
- Charts placed below for deeper analysis.
- Maps used for geographical insights.
- Consistent font, color scheme, icon style across all pages.
- Navigation tabs used for a smooth user journey.

✓ Interactive Elements

- **Slicers** used:
 - State Slicer (in State Analysis)
 - Constituency Slicer (in Constituency Analysis)
- **Drill-down** enabled on maps.
- **Tooltips** show additional details when hovering.

8.2 Page-Wise Visualization Design

Page 1: Overview Dashboard

Objective:

Provide a high-level national summary of the 2024 election results.

Visuals Used & Purpose

1. KPI Cards

- Total Seats (543)
- NDA Seats: 292 (54%)
- INDIA Seats: 234 (43%)
- Others: 17 (3%)

Purpose: Quick national-level understanding of seat distribution.

2. Alliance Summary Cards

- Same color pattern to differentiate alliances.
- Includes logos for visual identity.
- Shows % share to make comparison easier.

3. Alliance-wise Party Bar Sections

Each party placed with:

- Party logo
- Party name
- Total seats won

Purpose: Helps users see which party contributed to the alliance performance.

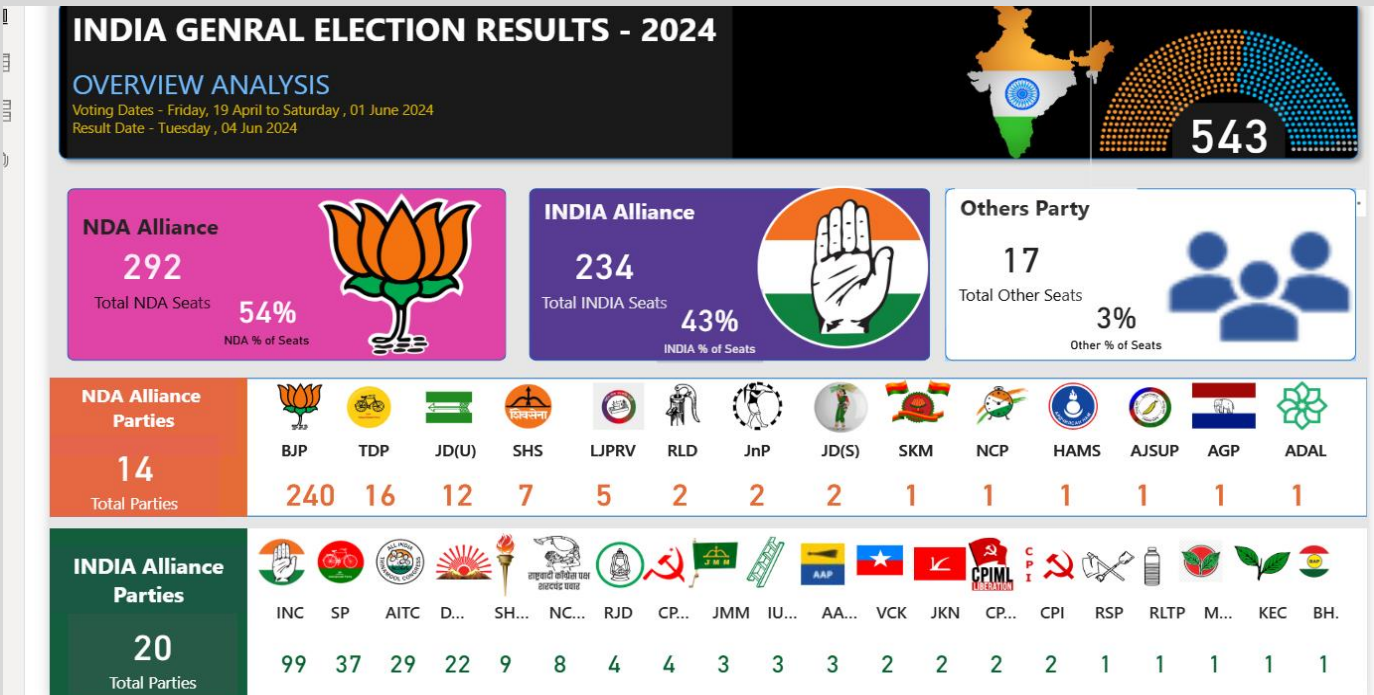


Fig no.1 Overview Analysis

Page 2: State Demographics Page

Objective:

Show state-wise election patterns and geographical trends.

Visuals Used

1. Filled Map – State Wise Seats

Shows:

- Total seats in each state
- Alliance with majority
- NDA vs INDIA seat count
- Color-coded alliance wins

Purpose: Identify regional strongholds of alliances.

2. Bubble Map – Constituency Winning Party

Each circle represents a constituency:

- Color = Alliance
- Larger bubble = More votes

Purpose: Shows micro-level distribution across the country.

3. State Majority Map

- **Blue** → INDIA majority
- **Saffron** → NDA majority

Purpose: Very clear visual representation of which alliance dominated each state.

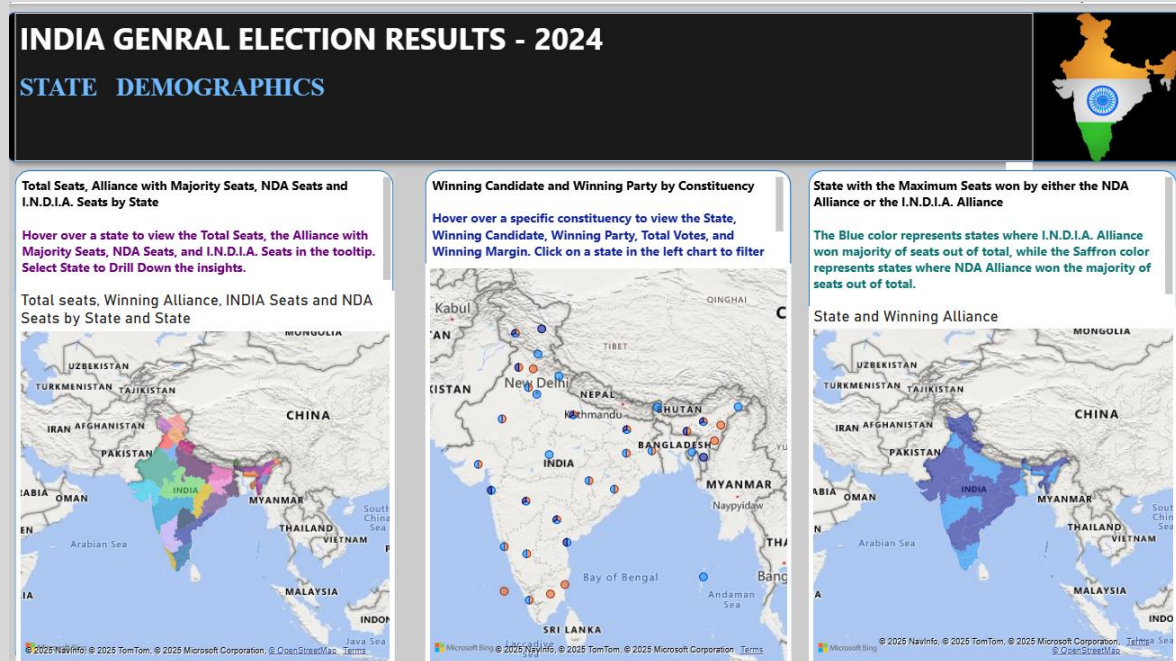


Fig no.2 State Demographics

Work Hours Summary

Activity	Hours Completed
Development	12 hours / 25 hours
Review	2 hours
Stand-up Meetings	1 hour

Conclusion

Milestone 2 successfully delivered:

- ✓ A clean data model
- ✓ Complete visualization design
- ✓ A functional Page 1 & Page 2 dashboard
- ✓ Core DAX measures
- ✓ Ready structure for final polished dashboard

The project is now prepared for Milestone 3: advanced DAX, formatting, interactions, and final presentation.

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Milestone 3 – Development Phase (Part 2 + Testing)

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Presented By: Shruti Bhore

1.Introduction

Milestone 3 focused on completing all remaining dashboard pages, adding interactive features, finalizing DAX measures, and beginning testing to ensure accuracy and usability.

Three major analytical pages were developed:

1. Overview Analysis Page
2. State Demographics Page
3. Constituency Analysis Page

These dashboards collectively provide a detailed, multi-level analysis of the 2024 Lok Sabha Election dataset.

2.Objectives

The primary objective of Milestone 3 is to complete the development of the analytical dashboard by enhancing its functionality, accuracy, and user experience. This milestone focuses on transforming the partially built pages into fully interactive and insight-driven reports. The key objectives include:

- Develop and finalize Pages 1–3 with focused analysis views.
- Complete advanced DAX measures required for insightful analytics.
- Add interactive features to improve user exploration and storytelling.
- Perform systematic testing (data accuracy, UI, performance).
- Prepare final review and packaging for presentation/deployment.

3.Work Completed in Milestone 3

3.1 Dashboard Page 1 – Overview Analysis

Objective:

To give a national-level summary of election performance by alliances and political parties.

Key Features Developed

- KPI Cards displaying:
 - Total Seats (543)
 - NDA Seats – 292 (54%)
 - INDIA Seats – 234 (43%)
 - Other Seats – 17 (3%)
- Semi-circular arc visual showing total Parliament seats.
- Alliance-wise summary with logos and percentages.
- Party-wise breakdown for NDA, INDIA, and OTHER groups.
- Interactive tooltips for party-wise performance.
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Value Added

- Enables users to understand the national results at a glance.
- Easy comparison of major alliances.
- Strong visual representation with clean color schemes.

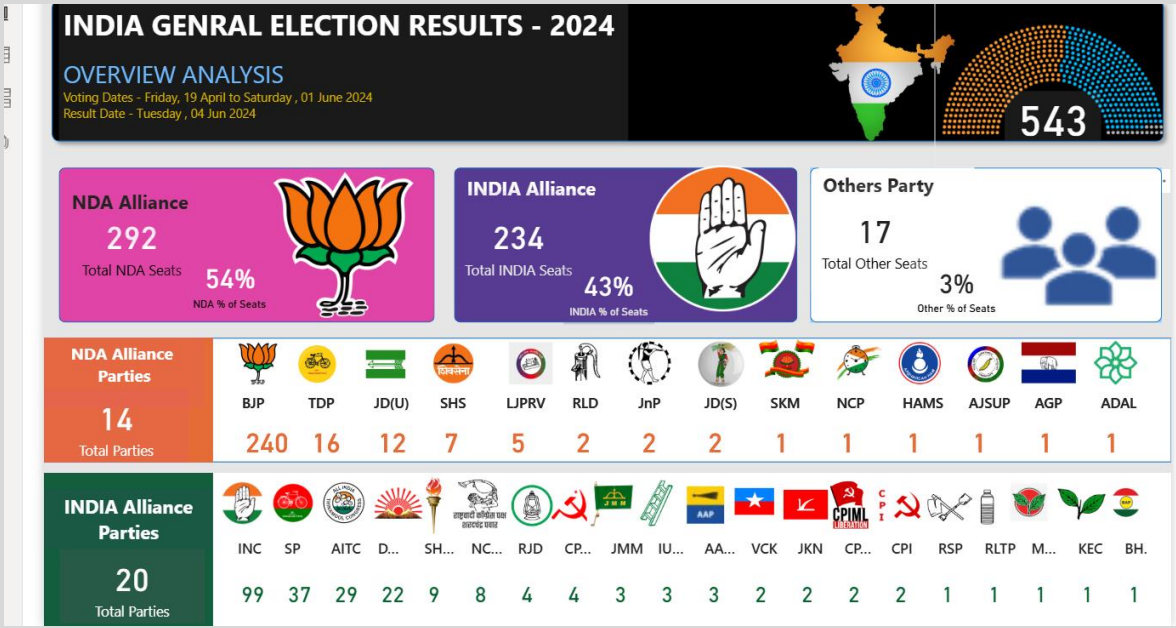


Fig no 1: Overview Analysis

3.2 Dashboard Page 2 – State Demographics

Objective:

To provide state-level and geographical insights using map-based analytics.

Key Visuals Implemented

1. State-wise Seat Distribution Map

- Shows total seats, NDA seats, and INDIA seats for each state.
- Hover interaction reveals detailed seat data
- .

2. Constituency and Winning Candidate Map

- Displays winning party, candidate, total votes, and margin for each constituency.
- Supports cross-filtering when a state is selected.

3. State-wise Winning Alliance Map

- Saffron color → NDA-majority states
- Blue color → INDIA-majority states
- Helps identify regional strongholds.

Value Added

- Provides spatial understanding of voting patterns.
- Shows political dominance at a state level.
- Offers drill-down access to constituency insights via map interaction.

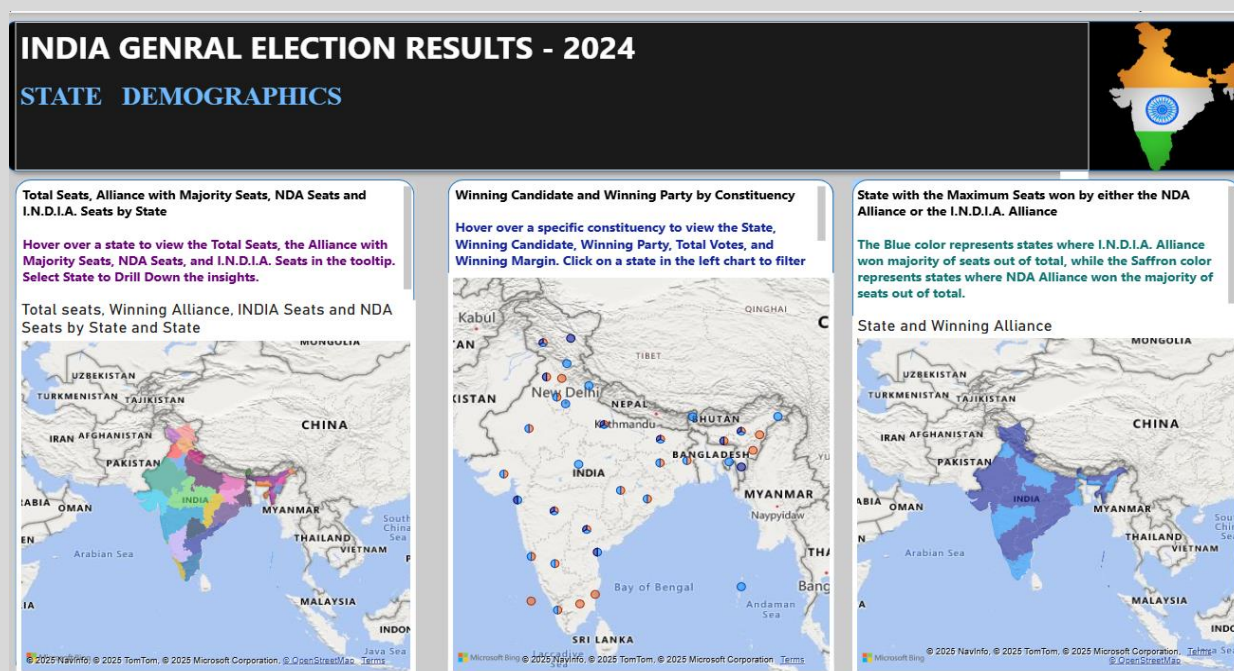


Fig no 2: State Demographics

3.3 Dashboard Page 3 – Constituency Analysis

Objective:

To create a micro-level analytical page for detailed constituency-based insights.

Key Features Developed

- Dropdown filter for selecting any constituency.
- **KPI cards showing:**
 - Total Votes
 - EVM Votes
 - Postal Votes
 - Total Candidates Participated
- **Detailed candidate cards:**
 - Winner: Name, Party, Vote Share, Total Votes
 - Runner-up: Name, Party, Vote Share
 - 2nd Runner-up: Name, Party, Vote Share
- Color-coded cards for better readability.

Value Added

- Provides a detailed breakdown of election results for each constituency.
- Helps understand voting patterns, competition levels, and margin of victory.
- Supports decision-making for political analysts and researchers.

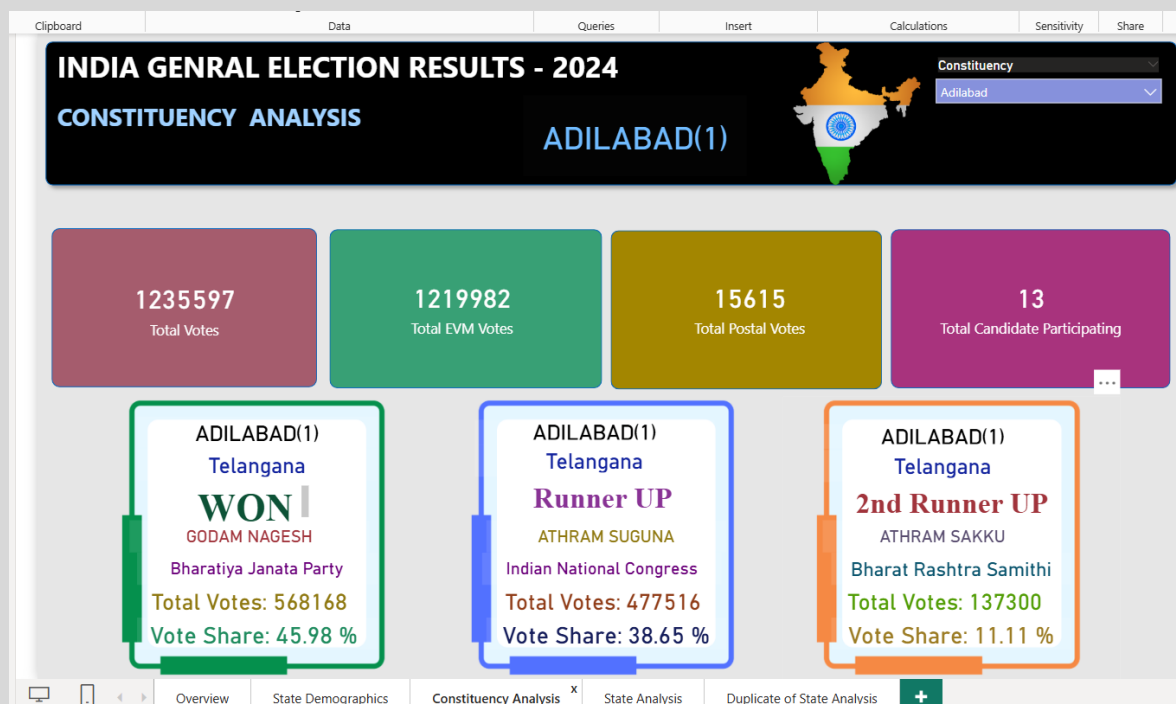


Fig no 3: Constituency Analysis

4.Interactivity Added

During Milestone 3, major interactivity features were implemented:

✓ Cross-Filtering Between Visuals

- Clicking on a state in the map instantly updates the other visuals (party seats, alliance comparison, demographics).
- Constituency markers on maps dynamically filter candidate details and vote counts.
- Alliance cards, party logos, and chart elements also act as filters for deeper exploration.

✓ Slicers for Dynamic Filtering

- State Slicer allows users to filter visuals to a specific state, enabling state-level analysis of seats and party performance.
- Constituency Slicer enables detailed drill-down to individual constituency results, including winners, runners-up, total votes, and vote share.
- Filters automatically update all visuals on the page, maintaining data consistency.

✓ Drill-Down & Hierarchical Navigation

- Users can move from National → State → Constituency levels through drill-down functionality.
- Enables multi-layered exploration while keeping the report structure clean.

✓ Tooltip-Based Insights

- Hovering over states or constituencies displays detailed information (such as total seats, winning party, winning margin, total votes).
- Tooltips help deliver insights without cluttering the dashboard visuals.

✓ Live page navigation between all 3 dashboards

This significantly improves user experience and analytical depth.

❖ Result

These interactive features significantly improve user engagement and make the dashboard more analytical, flexible, and decision-supportive. Users can derive insights instantly by interacting with any visual element.

5.DAX Measures Completed

All required DAX calculations were completed in this milestone, including:

- Total Seats
- Alliance-wise Seat Count
- Vote Share (%)
- Total Votes
- Margin Calculation
- Filtering Measures for Map Visuals

These measures support accurate, dynamic reporting across all dashboards.

6.Testing Performed

During Milestone 3, detailed testing was conducted to ensure that the dashboards were accurate, interactive, and user-ready. The testing focused on validating both the data accuracy and the functionality of all interactive components across the three pages—Overview, State Demographics, and Constituency Analysis.

Testing was conducted to ensure dashboard accuracy and performance:

✓ Functional Testing

- All visuals respond correctly to slicers.
- Maps filter properly by state and constituency.
- KPIs display accurate totals.

✓ Data Validation Testing

- Cross-checked seat counts and vote totals against raw dataset.
- Cross-checked all numerical values such as total seats, vote counts, and vote shares.
- Verified correctness of DAX measures used to calculate totals, percentages, and rankings.
- Ensured no mismatch between table data and visual outputs.

✓ Usability Testing

- Verified navigation flow between pages.
- Checked readability of cards and maps.

✓ Performance Testing

- Verified that dashboards load quickly without lag.
- Ensured optimized DAX calculations to avoid performance issues

Conclusion

Milestone 3 successfully delivered the final interactive dashboards of the Election Analysis project. All three pages—Overview, State Demographics, and Constituency Analysis—are fully functional, visually polished, and supported with accurate DAX calculations. During this milestone, significant progress was achieved in enhancing interactivity, refining the user experience, and ensuring data accuracy through comprehensive DAX calculations. The incorporation of slicers, drill-down features, cross-filtering, and dynamic maps has made the dashboard highly intuitive and user-friendly. Each visual element is backed by validated data and robust DAX measures, ensuring reliable and meaningful insights for end users. This milestone marks the near completion of the analytical development phase, with dashboards ready for final integration and review.